

ASJITH A MOOSA

SENIOR ENGINEER

Ernakulam, Kerala, India | Phone: +918156929684 | Email: asjithmuniparambil@gmail.com

LinkedIn: <https://www.linkedin.com/in/asjithamoosa/>

PROFESSIONAL SUMMARY

Highly motivated and results-driven Metallurgical and Materials Engineer with hands-on experience in casting, heat treatment, failure analysis, and quality control. Currently serving as a Senior Engineer at India's largest alloy wheel manufacturer, applying expertise in process optimization, metallography, and defect analysis. Proficient in industry tools such as spectrometry, hardness testing, and critical problem-solving. A strategic thinker with strong adaptability, team collaboration skills, and a passion for innovation in materials technology.

CORE COMPETENCIES

- Heat Treatment Processes (Tempering, Hardening, Normalizing)
- Metallurgical Analysis
- Non-Destructive Testing (X-ray)
- Casting Techniques (GDC, Flow Forming)
- Quality Control & Spectrometric Analysis
- RCA : QPR, Pareto, Fishbone, Why-Why Analysis
- Continuous Improvement (Kaizen, 5S, Gemba)

PROFESSIONAL EXPERIENCE

Senior Engineer - Casting

UNO MINDA Pvt Ltd. Gujarat | Minda Kosei Aluminum Wheels Pvt Ltd. Gujarat | Aug 2022 - Present

- Maintain and prepare IQPR, QC sheets, QCC, and documentation in line with process audits.
 - Collaborated with cross-functional teams to execute process improvements and reduce defects.
 - Utilized statistical process control (SPC) and data analysis to identify trends and ensure product quality.
 - Designed and optimized die casting processes to produce high-integrity aluminum alloy wheels.
 - Collaborated with tooling engineers to design and modify dies, ensuring dimensional accuracy and reducing production downtime.
 - Implemented lean manufacturing principles, reducing material waste by 25% and improved the production efficiency.
 - Performed root cause analysis on casting defects (e.g., porosity, shrinkage) and applied corrective actions to enhance product quality.
 - Monitored and controlled die casting parameters (e.g., pressure, temperature, cycle time) to ensure consistent product quality.
 - Conducted metallurgical analysis to select appropriate alloys and heat treatment processes for specific application.
 - Handled the chemical composition checking and hardness of the alloy.
 - Enforced various kaizens to optimize the current system.
 - Developed and sustained process documentation, including work instructions and standard operating procedures (SOPs).
 - Trained and mentored junior engineers and technicians on die casting best practices and safety protocols.
 - Collaborated with quality assurance teams to ensure compliance with industry standards.
- Performed SAP related works such as production punching, rework order management etc.
- Trained operators to understand MARU A activities and upheld 5S in the shop floor.

EDUCATION

PG Diploma - Oil and Gas Piping and Pipeline

National Skill Development Corporation, Kochi
India International Skill Centre | 2025

B.Tech - Metallurgical and Materials Engineering

Amal Jyothi College of Engineering, Kottayam
APJ Abdul Kalam Technological University | 2018 – 2022

ADDITIONAL QUALIFICATIONS

- ASNT NDT LEVEL II Certification
- Certified QA/QC
- Certified Elementary First Aider

CERTIFICATIONS & TRAINING

- PG Diploma in Oil and Gas Piping and Pipeline, March 2025 - July 2025
- NPTEL - Introduction to Materials Science & Engineering – IIT Madras
- NPTEL - Theory & Practice of Non-Destructive Testing – IIT Madras
- Internship - Embedded Systems - Pace Lab, Kochi

PROJECTS AND SEMINAR

- **Reduced Rejection Ratio from 3.5% to 1.3% (2025):** by maintaining the cooling and melt temperature, giving for awareness to manpower.
- **Increased per hour Production (2024) :** By identifying and eliminating bottlenecks, improving metal flow, accurate temperature control, increased inspection.
- **Reduction of Cycle Time in Casting process (2023):** reduced the cycle time in casting by forming CFT, deeply analyzing and installing new various methods.
- **Reduction in Melting Gas Consumption(2022-2023) :** by optimizing air-fuel ratio, improving insulation, process optimization, controlling parameters.
- **Effect of Cryogenic Treatment on Mechanical Properties of Ferritic Stainless Steel SS430 (2021-2022) :** the primary objective is to analyze changes in hardness and tensile strength post-cryogenic treatment using metallography, tensile testing.
- **Biomimetic Scaffolding in Bone Tissue Engineering (2021) :** the objective of this seminar is to create artificial bone tissue that mimics the natural bone tissue's matrix and structure.
- **Green Energy Fitness Machine (Piezoelectric Concept) (2020):** the objective of this is to develop a prototype for energy harvesting from mechanical movement.

TECHNICAL SKILLS

- Damage Mechanisms
- Corrosion Control and Monitoring
- Metallurgical Testing: Spectrometry Analysis, Microstructure Analysis
- Defect Analysis: Mechanical Testing, NDT
- Software: SAP, MS Office (Excel, PowerPoint, Word)

LANGUAGES

- English
- Malayalam
- Tamil
- Hindi